An introduction to Bayesian modelling with brms and Stan

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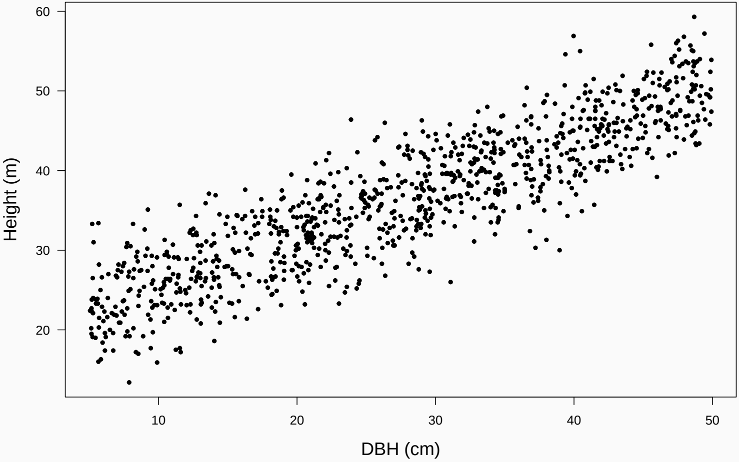
Our dataset: tree heights and DBH

- One species
- 10 plots
- 1000 trees
- Number of trees per plot ranging from 4 to 392

```
trees <- read.csv('data/trees.csv')
```

```
      site              dbh               height     
 Min.   : 1.0   Min.   : 5.06   Min.   :13.40  
 1st Qu.: 1.0   1st Qu.:17.69   1st Qu.:29.68  
 Median : 2.0   Median :28.62   Median :36.55  
 Mean   : 2.7   Mean   :27.88   Mean   :36.51  
 3rd Qu.: 4.0   3rd Qu.:38.97   3rd Qu.:43.33  
 Max.   :10.0   Max.   :49.92   Max.   :59.30  
```

What’s the relationship between DBH and height?



First step: linear regression (lm)

```r
simple.lm <- lm(height ~ dbh, data = trees)
```

```
Call:
lm(formula = height ~ dbh, data = trees)

Residuals:
     Min       1Q   Median       3Q      Max
-13.3270  -2.8978   0.1057   2.7924  12.9511

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 19.33920    0.31064   62.26   <2e-16 ***
dbh          0.61570    0.01013   60.79   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.093 on 998 degrees of freedom
Multiple R-squared:  0.7874,    Adjusted R-squared:  0.7871
F-statistic:  3695 on 1 and 998 DF,  p-value: < 2.2e-16
```

Center continuous variables

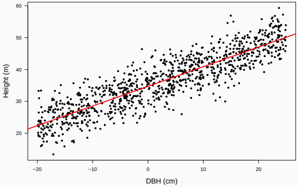
```r
summary(trees$dbh)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
   5.06   17.69   28.62   27.88   38.97   49.92 
```

```r
trees$dbh.c <- trees$dbh - 25
```

So, all parameters will be referred to a 25 cm DBH tree.

Linear regression with centred DBH

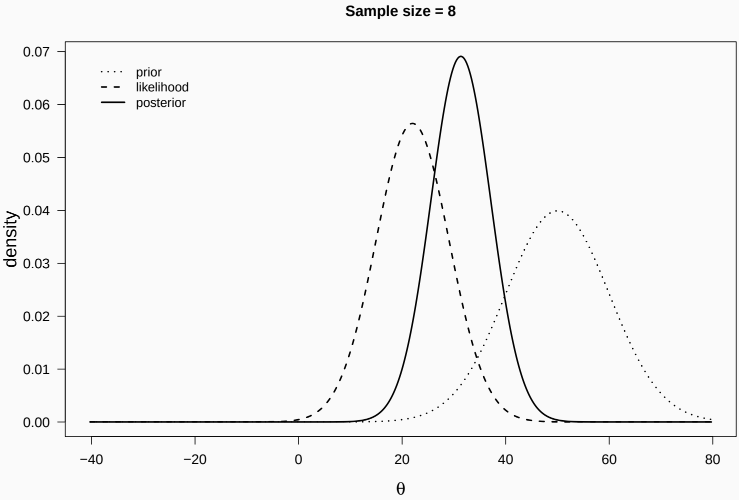



```
lm(formula = height ~ dbh.c, data = trees
            coef.est coef.se
(Intercept) 34.73    0.13
dbh.c        0.62    0.01
---
n = 1000, k = 2
residual sd = 4.09, R-Squared = 0.79
```

Let's make it Bayesian

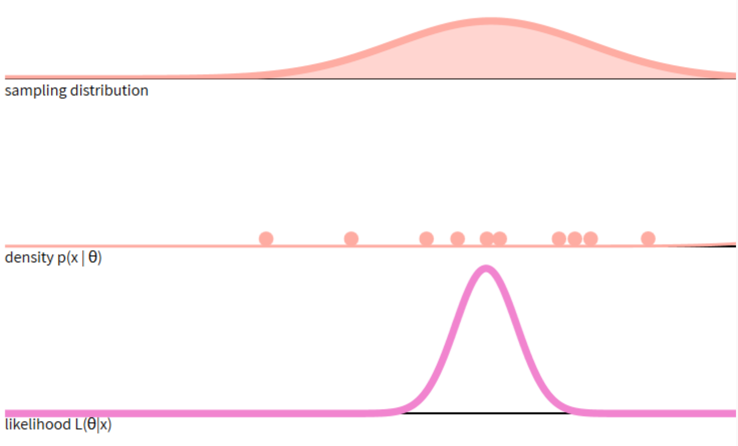
$$P(Unknown|Data) \propto P(Data|Unknown) \times P(Unknown)$$

$$Posterior \propto Likelihood \times Prior$$



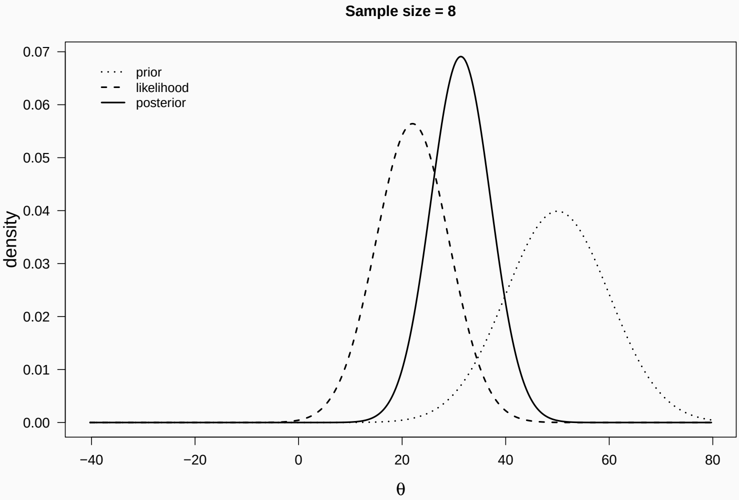
What is the likelihood?

$$L(\theta|x) = P(x|\theta)$$

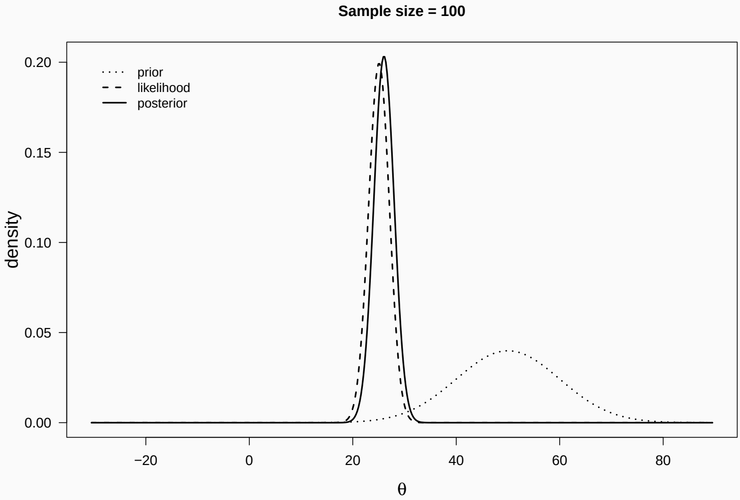


https://seeing-theory.brown.edu/bayesian-inference/index.html

Bayesian inference: prior and likelihood produce posterior



With increasing sample size, likelihood dominates prior



Bayesian statistics in practice

- Integrate information (**prior**)

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- **Uncertainty** / **Propagate errors**

Remember our model structure

$$y_i \sim N(\mu_i, \sigma^2)$$
$$\mu_i = \alpha + \beta x_i$$

In this case:

$$Height_i \sim N(\mu_i, \sigma^2)$$
$$\mu_i = \alpha + \beta DBH_i$$

α: expected height when DBH = 25 cm

β: how much height increases with every unit increase of DBH

Defining model formula

```
library('brms')

height.formu <- brmsformula(height ~ dbh.c)
```

We must define **prior distributions** for every parameter

brms default priors

```r
get_prior(height.formu, data = trees)
```

```
                 prior     class  coef group resp dpar nlpar lb ub
                (flat)         b
                (flat)         b dbh.c
 student_t(3, 36.5, 10.2) Intercept
   student_t(3, 0, 10.2)     sigma                              0
       source
      default
 (vectorized)
      default
      default
```

Choosing priors

Avoid ‘non-informative’ priors

Use *weakly informative* (e.g. relatively wide Normal or t-student distributions)

or *strongly informative* priors based on previous knowledge and common sense.

Some tips for setting priors:

- https://github.com/stan-dev/stan/wiki/Prior-Choice-Recommendations

Run **prior predictive checks** (just priors, no data)

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- `https://doi.org/10.1111/oik.05985`

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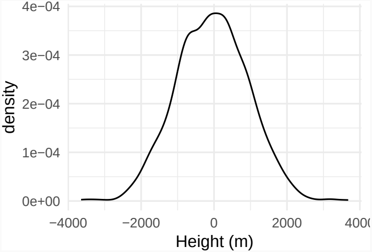
Probability distribution explorer

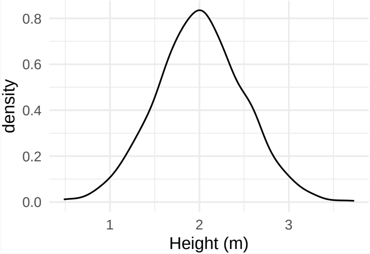
https://distribution-explorer.github.io/

Example: estimating people height across countries

Unreasonable prior



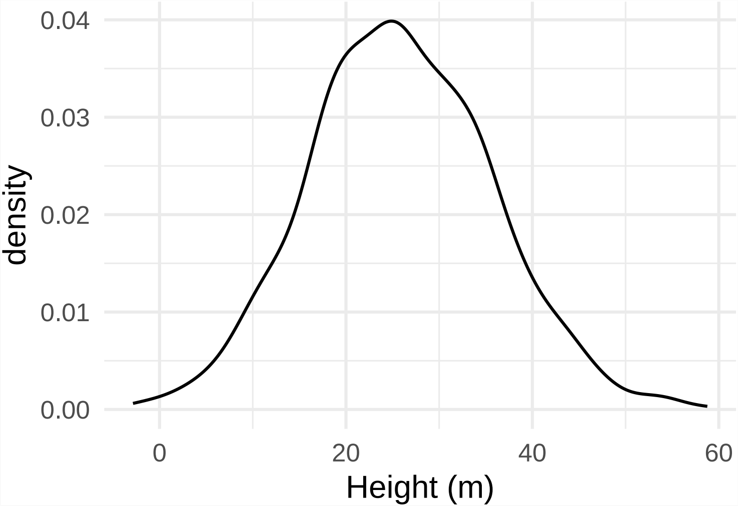
Reasonable prior



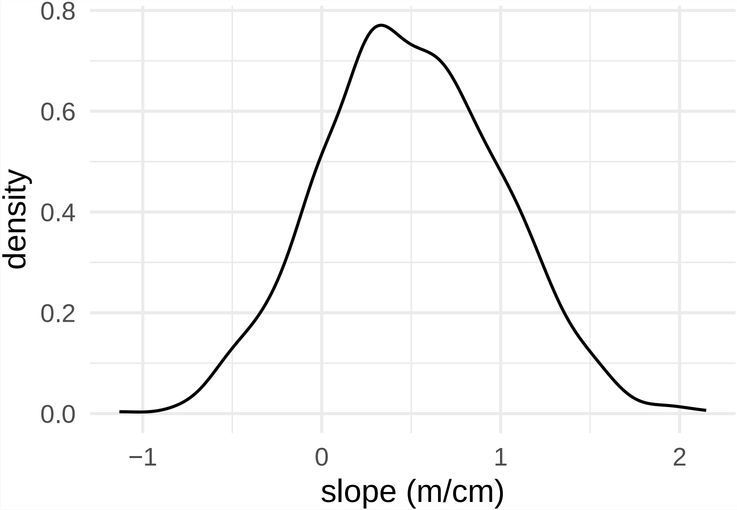
Defining priors for our trees example

```r
priors <- c(
  set_prior('normal(30, 10)', class = 'Intercept'),
  set_prior('normal(0.5, 0.4)', class = 'b'),
  set_prior('normal(0, 5)', class = 'sigma')
)
```

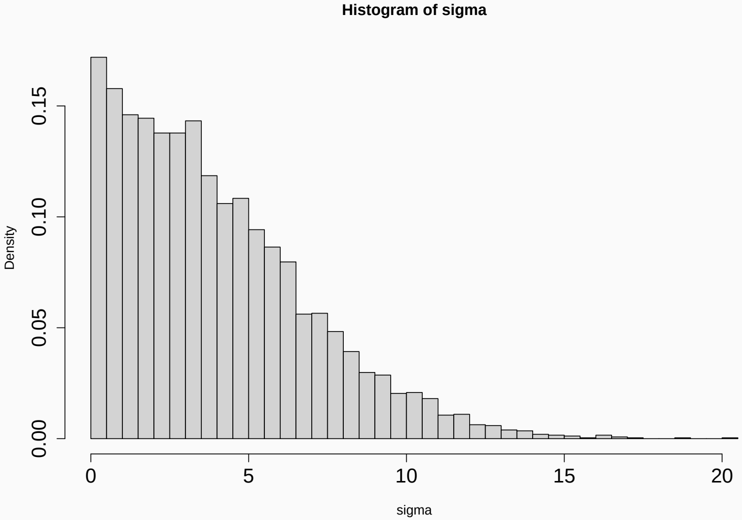
Prior for intercept (average height of 25-cm diameter tree)



Prior for slope



Prior for sigma (residual sd)

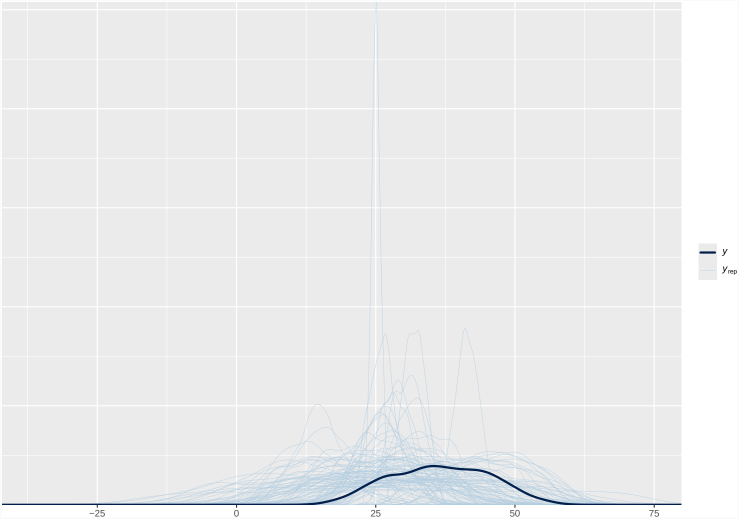


Prior predictive check

```r
height.mod <- brm(height.formu,
         data = trees,
         prior = priors,
         sample_prior = 'only')
```

Prior predictive check

```r
pp_check(height.mod, ndraws = 100)
```



Fit model (now with data)

```
height.mod <- brm(height.formu,
         data = trees,
         prior = priors)
```

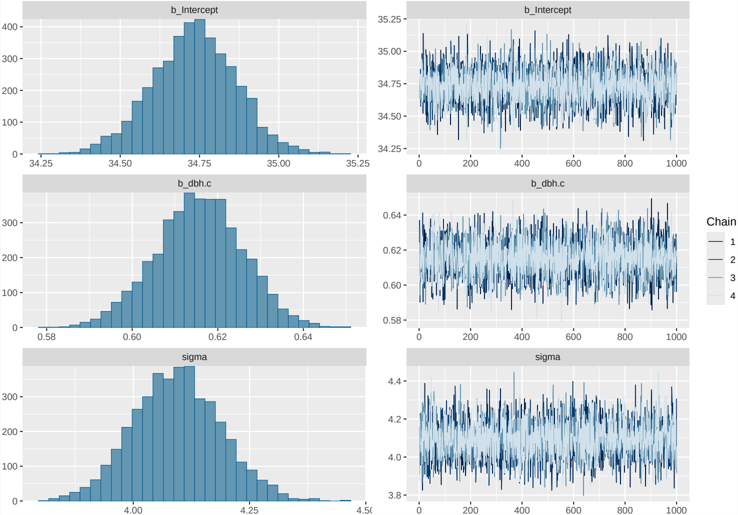
Model summary

```r
summary(height.mod)
```

```
 Family: gaussian
  Links: mu = identity; sigma = identity
Formula: height ~ dbh.c
   Data: trees (Number of observations: 1000)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept    34.73      0.13    34.47    34.99 1.00     4222     3247
dbh.c         0.62      0.01     0.60     0.64 1.00     4499     2673

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     4.09      0.09     3.92     4.28 1.00     4430     3153

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

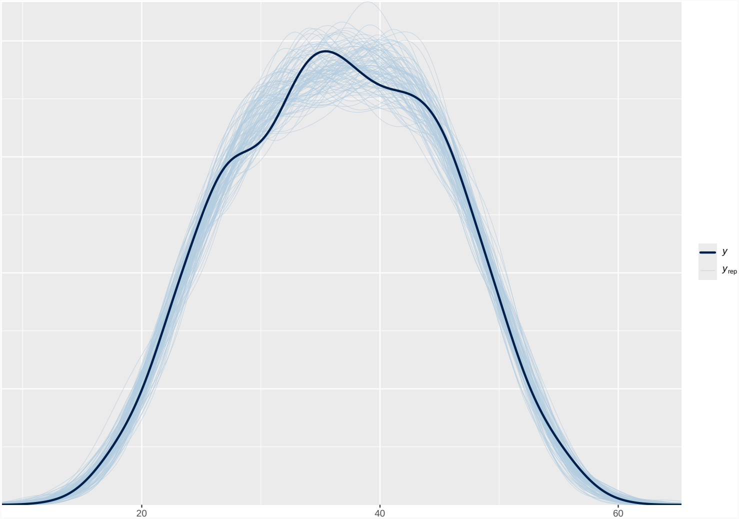
Model visualisation

```r
plot(height.mod)
```



Posterior predictive checking

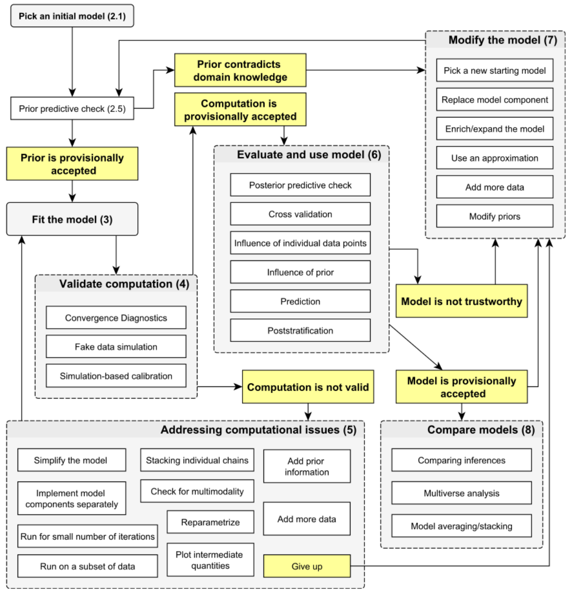
```r
pp_check(height.mod, ndraws = 100)
```



Interactive model exploration

```r
library('shinystan')
launch_shinystan(height.mod)
```

The Bayesian workflow



https://arxiv.org/abs/2011.01808

Exercise

height ~ sex

To read more

Regression and other stories

Statistical Rethinking

Statistical rethinking with brms, ggplot2, and the tidyverse

Bayesian Population Analysis using WinBugs

Applied Hierarchical Modeling in Ecology

Stan user guide